CMPG 322 - Decision Support Systems (DSS) 2
SU5 - Project Management



Mr. JH (Heystek) Grobler

School of Computer Science and Information Systems
Faculty of Natural and Agricultural Sciences
North-West University (NWU)

Outline

- How to plan, monitor, and control projects with the use of PERT and CPM;
- Earliest start, earliest finish, latest start, latest finish, and slack times for each activity, total project completion time and total project cost;
- Reducing total project time at the least total cost by crashing the network;
- Software in project management.

- Managing large-scale, complicated projects effectively is a difficult problem and the stakes are high
  - The first step in planning and scheduling a project is to develop the **work breakdown structure (WBS)**
  - Identify **activities** that must be performed and their beginning and ending **events**
  - Identify time, cost, resource requirements, predecessors, and people responsible for each activity
  - A schedule for the project then can be developed

- The **program evaluation and review technique (PERT) and the critical path method (CPM)** are two popular quantitative analysis techniques for complex projects
  - PERT uses three time estimates to develop a probabilistic estimate of completion time
  - CPM is a more deterministic technique
  - They are so similar they are commonly considered as one technique, PERT/CPM

Six steps of PERT/CPM

1. Define the project and all of its significant activities or tasks.
2. Develop the relationships among the activities. Decide which activities must precede others.
3. Draw the network connecting all of the activities.
4. Assign time and/or cost estimates to each activity.
5. Compute the longest time path through the network; this is called the critical path.
6. Use the network to help plan, schedule, monitor, and control the project.

1. When will the entire project be completed?
2. What are the critical activities or tasks in the project–that is, the ones that will delay the entire project if they are late?
3. Which are the noncritical activities –that is, the ones that can run late without delaying the entire project's completion?
4. If there are three time estimates, what is the probability that the project will be completed by a specific date?
5. At any particular date, is the project on schedule, behind schedule, or ahead of schedule?
6. On any given date, is the money spent equal to, less than, or greater than the budgeted amount?
7. Are there enough resources available to finish the project on time?

General Foundry Example of PERT/CPM

General Foundry, Inc., a metalworks plant in Milwaukee, has long been trying to avoid the expense of installing air pollution control equipment. The local environmental protection group has recently given the foundry 16 weeks to install a complex air filter system on its main smokestack. General Foundry was warned that it will be forced to close unless the device is installed in the allotted period. Lester Harky, the managing partner, wants to make sure that installation of the filtering system progresses smoothly and on time.

ACTIVITY	DESCRIPTION	IMMEDIATE PREDECESSORS
A	Build internal components	–
B	Modify roof and floor	–
C	Construct collection stack	A
D	Pour concrete and install frame	B
E	Build high-temperature burner	C
F	Install pollution control system	C
G	Install air pollution device	D, E
H	Inspect and test pollution control system	F, G

Table: Table 11.1 Activities and Immediate Predecessors for General Foundry, Inc.

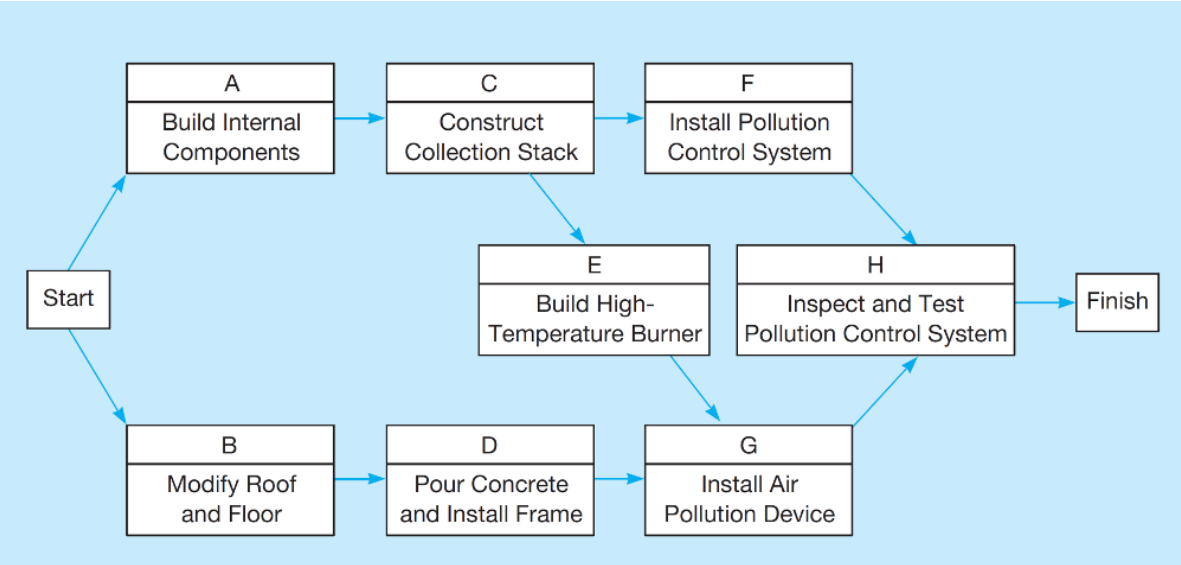
Drawing the PERT/CPM network

- **Activity-on-node (AON)** – nodes represent activities
- **Activity-on-arc (AOA)** – arcs represent the activities
- The AON approach is easier and more commonly found in software packages
- One node represents the start of the project, one node for the end of the project, and nodes for each of the activities
- The arcs are used to show the predecessors for each activity

Example: General Foundry

ACTIVITY	DESCRIPTION	IMMEDIATE PREDECESSORS
A	Build internal components	–
B	Modify roof and floor	–
C	Construct collection stack	A
D	Pour concrete and install frame	B
E	Build high-temperature burner	C
F	Install pollution control system	C
G	Install air pollution device	D, E
H	Inspect and test pollution control system	F, G

Table: Table 11.1 Activities and Immediate Predecessors for General Foundry, Inc.

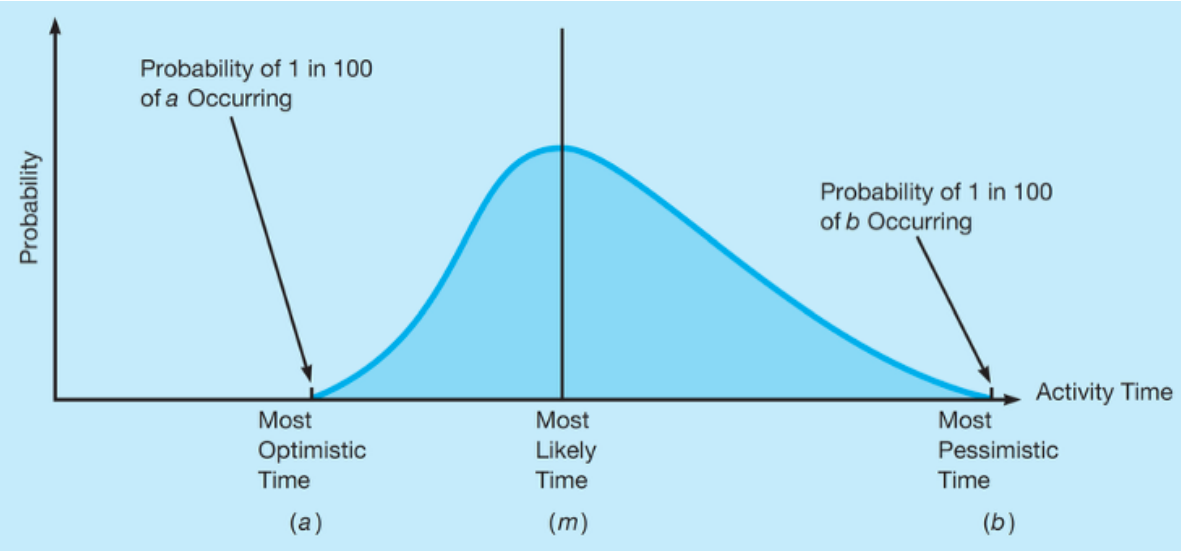


PERT often assumes time estimates follow a beta probability distribution

The time estimates in PERT are

- **Optimistic time (a)** = time an activity will take if everything goes as well as possible. There should be only a small probability (say, 1/100) of this occurring.
- **Pessimistic time (b)** = time an activity would take assuming very unfavorable conditions. There should also be only a small probability that the activity will really take this long.
- **Most likely time (m)** = most realistic time estimate to complete the activity.

FIGURE 11.2 Beta Probability Distribution with Three Time Estimates



- To find the **expected activity time** (t), the beta distribution weights the estimates as follows

$$t = \frac{a + 4m + b}{6}$$

- To compute the dispersion or **variance of activity completion time**

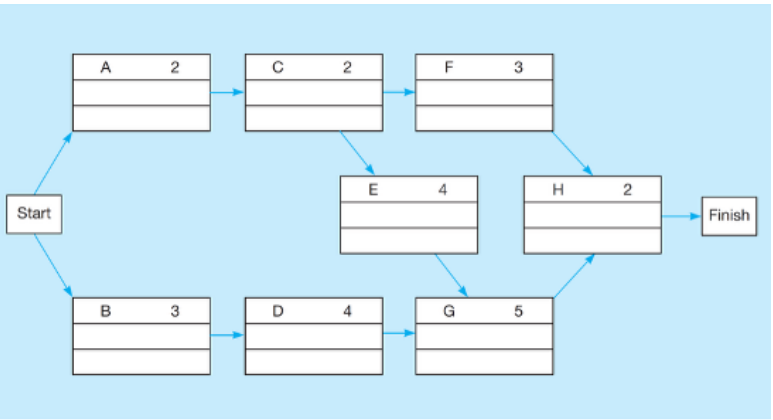
$$\text{Variance} = \left(\frac{b - a}{6}\right)^2$$

TABLE 11.2 Time Estimates (Weeks) for General Foundry, Inc.

Activity	**Optimistic,** a	**Most Likely,** m	**Pessimistic,** b	**Expected Time,** $t = \frac{a+4m+b}{6}$	**Variance,** $\left(\frac{b-a}{6}\right)^2$
A	1	2	3	2	0
B	2	3	4	3	0.11
C	1	2	3	2	0.11
D	2	4	6	4	0.44
E	1	4	7	4	1.00
F	1	2	9	3	1.78
G	3	4	11	5	1.78
H	1	2	3	2	0.11

FIGURE 11.3 General Foundry's Network with Expected Activity Times

ACTIVITY	EXPECTED TIME
A	2
B	3
C	2
D	4
E	4
F	3
G	5
H	2



To find the critical path, determine the following quantities for each activity:

1. **Earliest start (ES) time:** the earliest time an activity can begin without violation of immediate predecessor requirements
2. **Earliest finish (EF) time:** the earliest time at which an activity can end
3. **Latest start (LS) time:** the latest time an activity can begin without delaying the entire project
4. **Latest finish (LF) time:** the latest time an activity can end without delaying the entire project

- Activity times are represented in the nodes

Notation in node

ACTIVITY	t
ES	EF
LS	LF

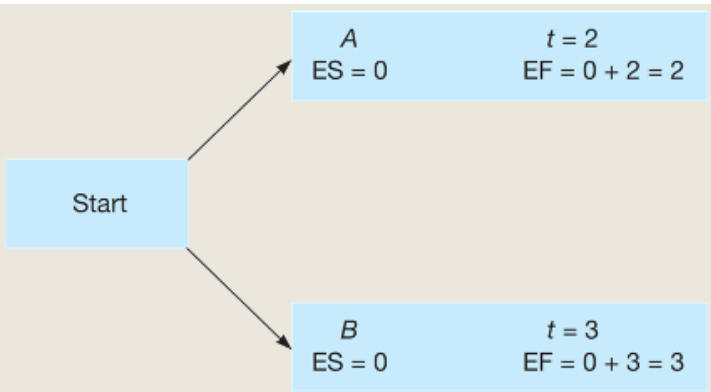
- Earliest finish time (EF) = Earliest start time (ES) + Expected activity time (t)

$$EF = ES + t$$

- Earliest start (ES) = Largest of the earliest finish times of immediate predecessors

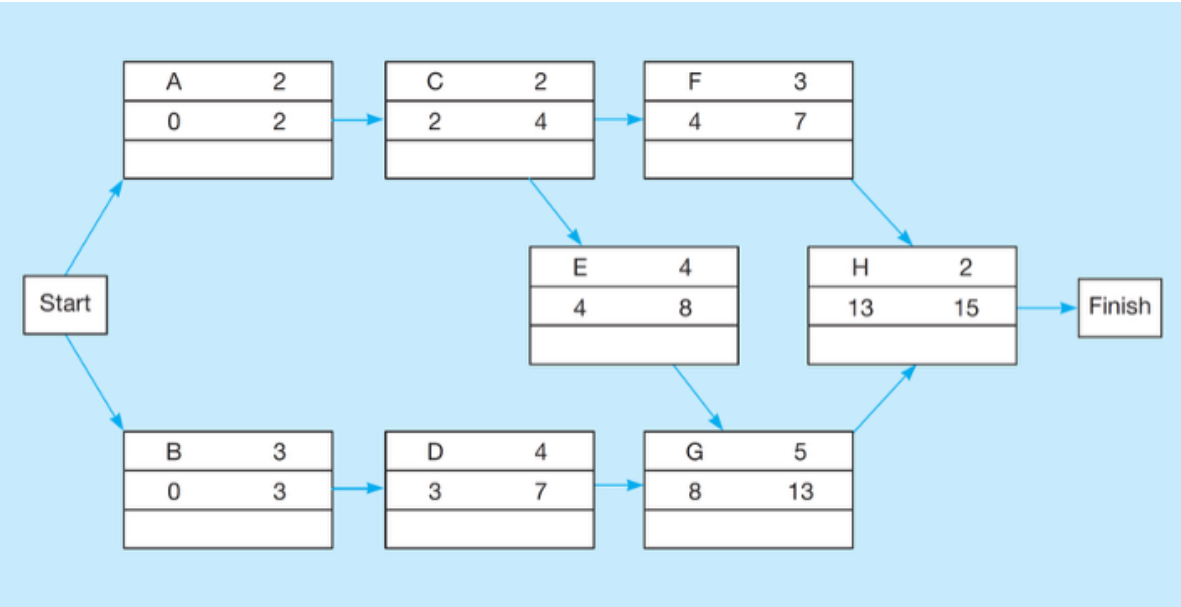
Critical path

- At the start of the project we set the time to zero
- Thus ES = 0 for both A and B



ACTIVITY	EXPECTED TIME
A	2
B	3
C	2
D	4
E	4
F	3
G	5
H	2

FIGURE 11.4 General Foundry's Earliest Start (ES) and Earliest Finish (EF) Times



- Latest start time (LS) = Latest finish time (LF) – Expected activity time (t)

$$LS = LF - t$$

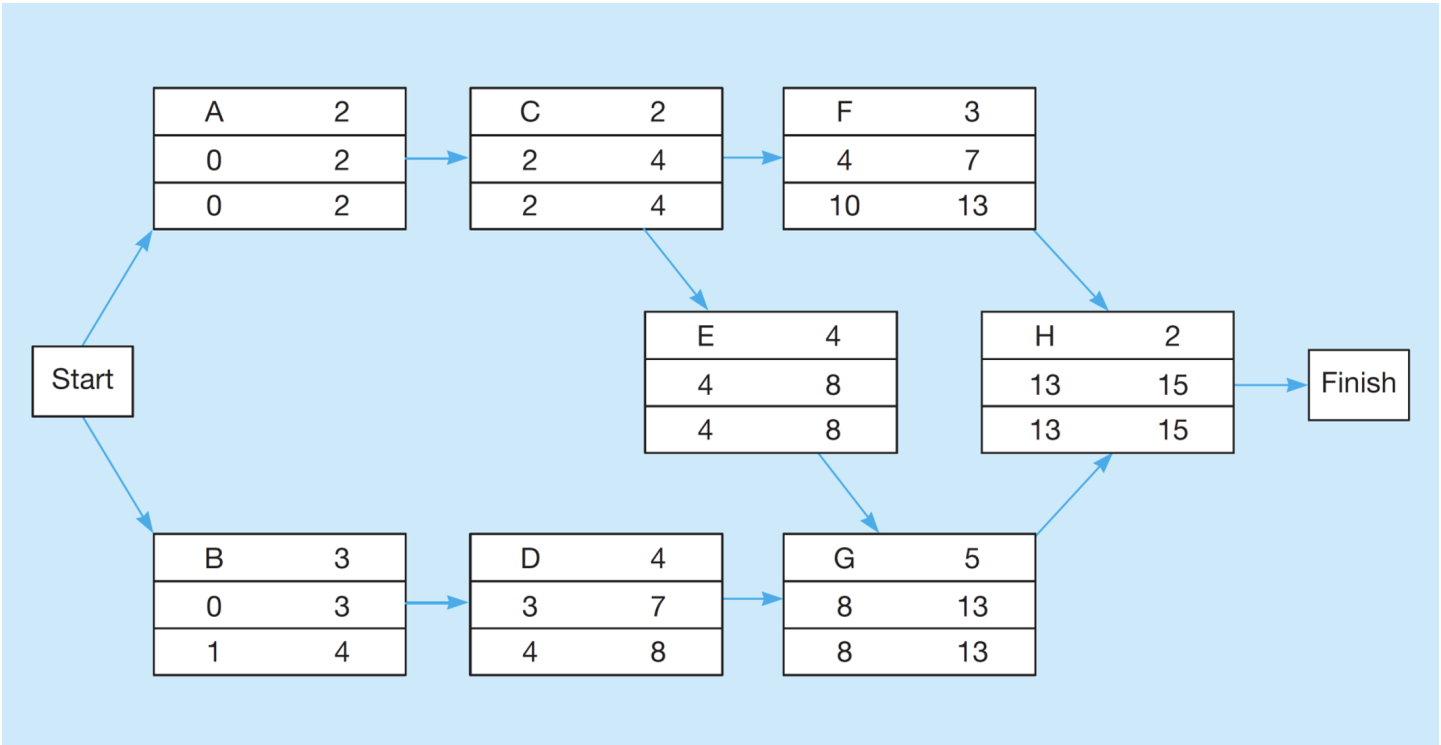
- Latest finish time (LF) = Smallest of latest start times for following activities

For activity H

$$LS = LF - t = 15 - 2 = 13 \text{ weeks}$$

$$LF = \text{Smallest LS of following activities}$$

FIGURE 11.5 General Foundry's Latest Start (LS) and Latest Finish (LF) Times



Critical path

- Amount of slack time for each activity

$$\text{Slack} = LS - ES \quad \text{or} \quad \text{Slack} = LF - EF$$

- Activities A, C, E, G, and H have no slack time
- These are called critical activities and they are said to be on the critical path
- The total project completion time is 15 weeks
- Industrial managers call this a boundary timetable

Activity	ES	EF	LS	LF	Slack	On Critical Path?
A	0	2	0	2	0	Yes
B	0	3	1	4	1	No
C	2	4	2	4	0	Yes
D	3	7	4	8	1	No
E	4	8	4	8	0	Yes
F	4	7	10	13	6	No
G	8	13	8	13	0	Yes
H	13	15	13	15	0	Yes

FIGURE 11.6 General Foundry's Critical Path (A–C–E–G–H)

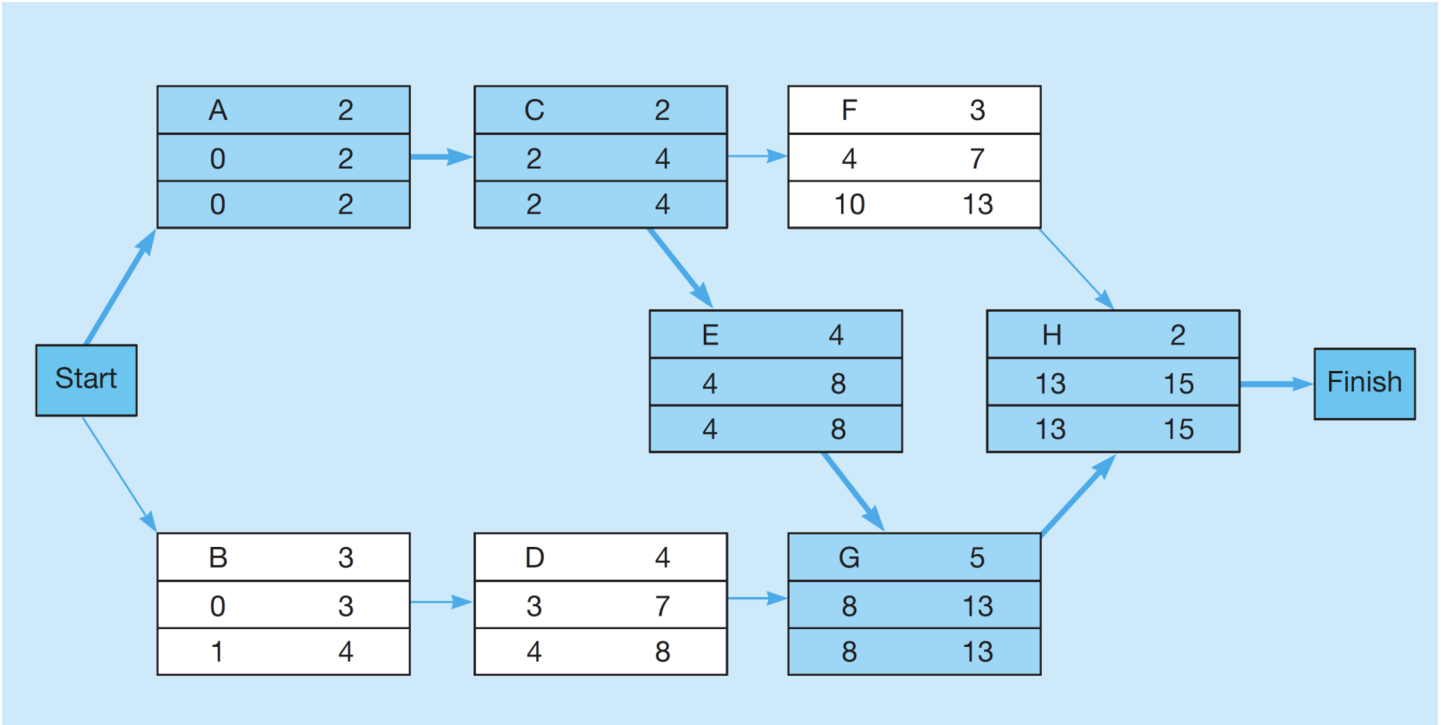
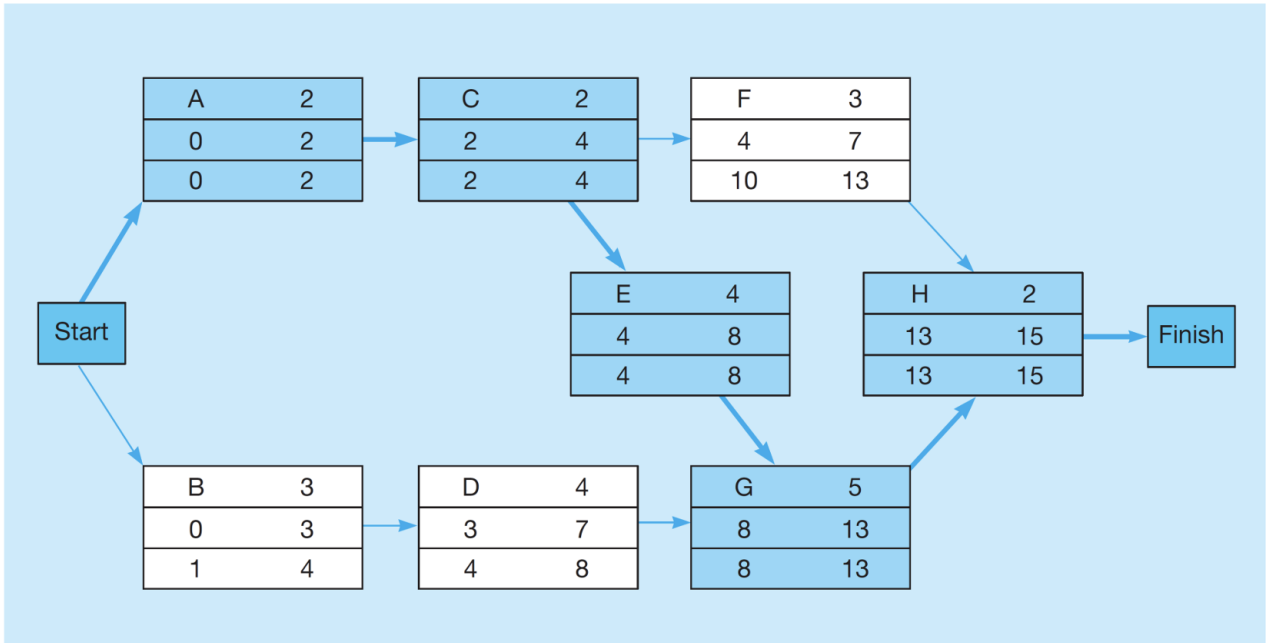


FIGURE 11.6 General Foundry's Critical Path (A–C–E–G–H)

General Foundry Example of PERT/CPM

General Foundry, Inc., a metalworks plant in Milwaukee, has long been trying to avoid the expense of installing air pollution control equipment. The local environmental protection group has recently given the foundry 16 weeks to install a complex air filter system on its main smokestack. General Foundry was warned that it will be forced to close unless the device is installed in the allotted period. Lester Harky, the managing partner, wants to make sure that installation of the filtering system progresses smoothly and on time.



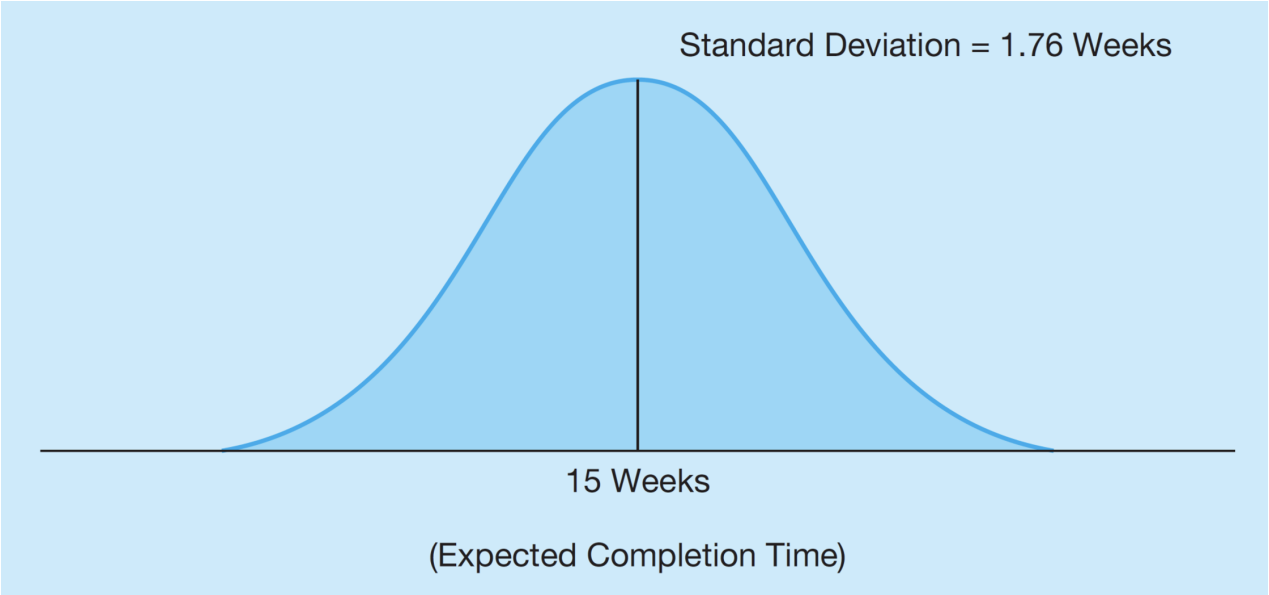
- PERT uses the variance of critical path activities to help determine the variance of the overall project:

$$\text{Project variance} = \sum \text{variances of activities on the critical path}$$

$$\text{Project variance} = \frac{36}{36} + \frac{64}{36} + \frac{4}{36} + \frac{4}{36} + \frac{112}{36} = \frac{112}{36} = 3.111$$

$$\text{Project standard deviation } \sigma_T = \sqrt{\text{Project variance}} = \sqrt{3.111} = 1.76 \text{ weeks}$$

FIGURE 11.7 Probability Distribution for Project Completion Times



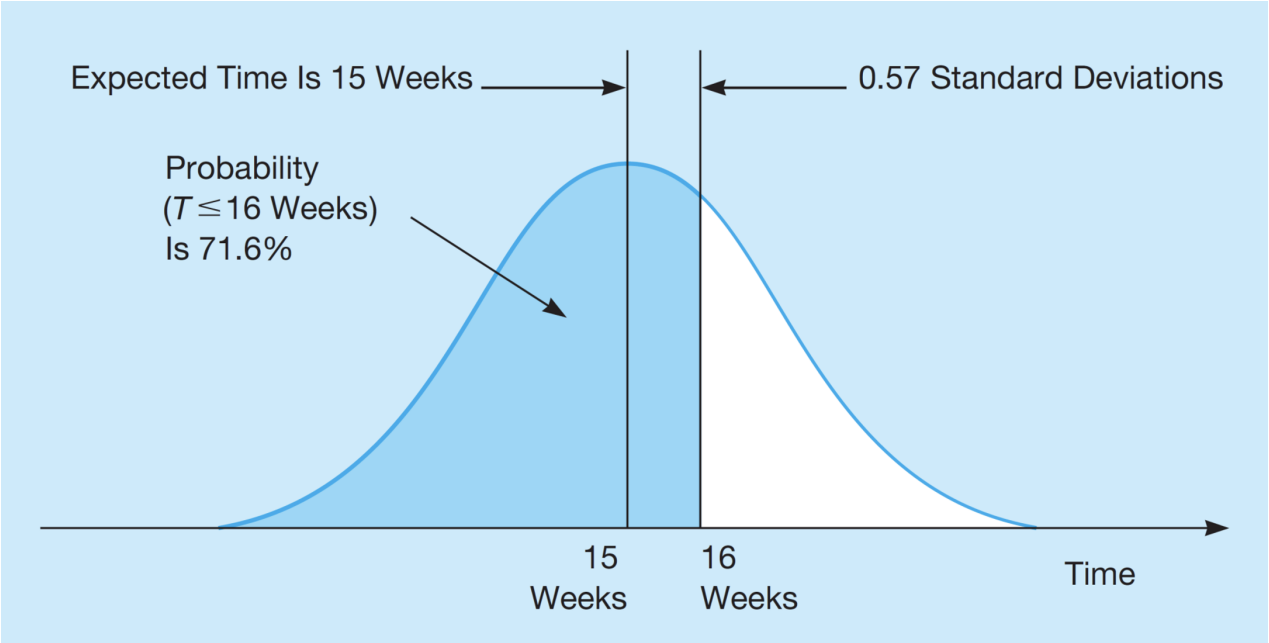
- The standard normal equation can be applied as follows:

$$Z = \frac{\text{Due date} - \text{Expected date of completion}}{\sigma_T}$$

$$Z = \frac{16 \text{ weeks} - 15 \text{ weeks}}{1.76 \text{ weeks}} = 0.57$$

- From Appendix A we find the probability of 0.71566 associated with this Z value
- That means the probability this project can be completed in 16 weeks or less is **0.716**

FIGURE 11.8 Probability of General Foundry's Meeting the 16-Week Deadline



What PERT was able to provide

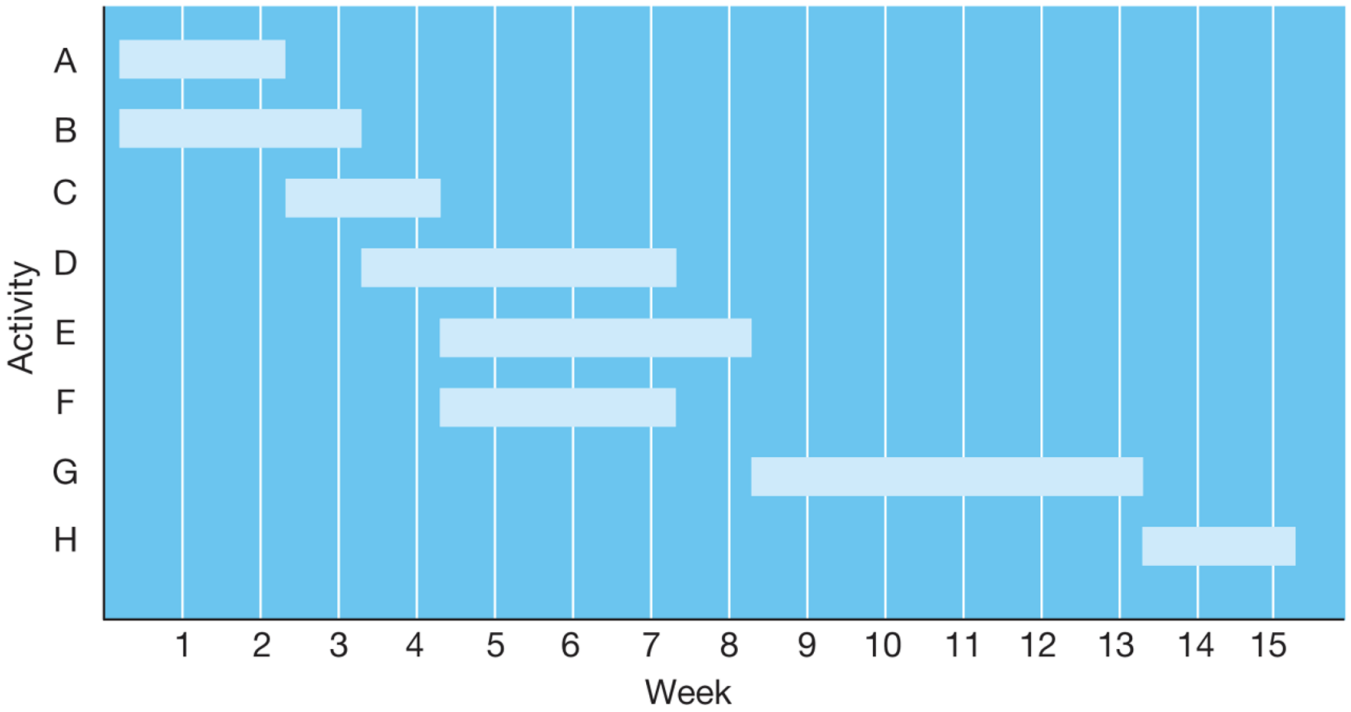
- The project's expected completion date is 15 weeks
- There is a 71.6% chance that the equipment will be in place within the 16-week deadline
- Five activities (A, C, E, G, H) are on the critical path
- Three activities (B, D, F) are not critical but have some slack time built in
- A detailed schedule of activity starting and ending dates has been made available

Four Steps of the Budgeting Process

1. Identify all costs associated with each of the activities. Then add these costs together to get one estimated cost or budget for each activity.
2. If you are dealing with a large project, several activities can be combined into larger work packages. A work *package* is simply a logical collection of activities. Since the General Foundry project we have been discussing is small, each activity will be a work package.
3. Convert the budgeted cost per activity into a cost per time period. To do this, we assume that the cost of completing any activity is spent at a uniform rate over time. Thus, if the budgeted cost for a given activity is \$48,000 and the activity's expected time is 4 weeks, the budgeted cost per week is \$12,000 (\$48,000/4 weeks).
4. Using the earliest and latest start times, find out how much money should be spent during each week or month to finish the project by the date desired.

Budgeting for General Foundry

- The Gantt chart below illustrates this process
- Determine how much will be spent on each activity during each week and fill these amounts into a chart in place of the bars



Activity	Earliest Start (ES)	Latest Start (LS)	Expected Time (t)	Total Budgeted Cost ($)	Budgeted Cost per Week ($)
A	0	0	2	22,000	11,000
B	0	1	3	30,000	10,000
C	2	2	2	26,000	13,000
D	3	4	4	48,000	12,000
E	4	4	4	56,000	14,000
F	4	10	3	30,000	10,000
G	8	8	5	80,000	16,000
H	13	13	2	16,000	8,000
Total				308,000	

Table: Activity Cost for General Foundry, Inc.

Budgeting for General Foundry

Activity	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total
A	11	11														22
B		10	10	10												30
C			13	13												26
D			12	12	12											48
E						14	14	14								42
F						10	10	10								30
G								16	16	16	16	16				80
H														8	8	16
Total per week	21	21	23	25	36	36	36	46	16	16	16	16	8	8		308
Total to date	21	42	65	90	126	162	198	244	260	276	292	308	292	300	308	308

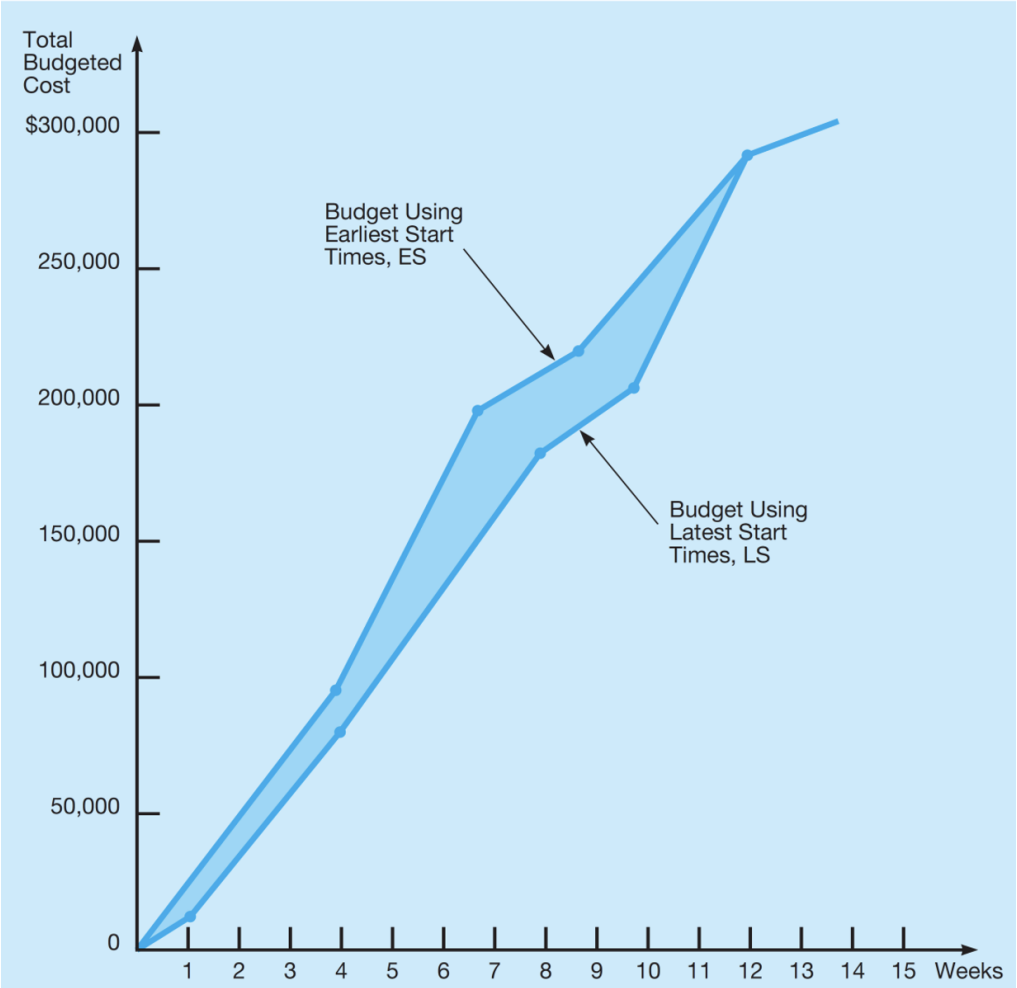
Table: Budgeted Cost ($1,000s) for General Foundry, Inc., Using Earliest Start Times

Budgeting for General Foundry

Activity	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total
A	11	11														22
B	10	10	10													30
C			13	13												26
D				12	12	12	12									48
E					14	14	14	14								56
F									10	10	10					30
G								16	16	16	16	16	16			80
H													8	8		16
Total per week	11	21	23	23	26	26	26	26	16	16	26	26	26	8	8	308
Total to date	11	32	55	78	104	130	156	182	198	214	240	266	292	300	308	

Table: Budgeted Cost ($1,000s) for General Foundry, Inc., Using Latest Start Times

FIGURE 11.10 Budget Ranges for General Foundry



- The value of work completed, or the cost to date for any activity
- The activity difference

$$\text{Value of work completed} = (\%\text{ of work complete}) \times (\text{Total activity budget})$$

$$\text{Activity difference} = \text{Actual cost} - \text{Value of work completed}$$

Activity	Total Budgeted Cost ($)	Percent of Completion	Value of Work Completed ($)	Actual Cost ($)	Activity Difference ($)
A	22,000	100%	22,000	20,000	$-2,000$
B	30,000	100%	30,000	36,000	$+6,000$
C	26,000	100%	26,000	26,000	0
D	48,000	10%	4,800	6,000	$+1,200$
E	56,000	20%	11,200	20,000	$+8,800$
F	30,000	20%	6,000	4,000	$-2,000$
G	80,000	0%	0	0	0
H	16,000	0%	0	0	0
Total	308,000		100,000	112,000	$+12,000$

Table: Monitoring and Controlling Budgeted Cost for General Foundry, Inc.

Activity	Total Budgeted Cost ($)	Percent of Completion	Value of Work Completed ($)	Actual Cost ($)	Activity Difference ($)
A	22,000	100%	22,000	20,000	$-2,000$
B	30,000	100%	30,000	36,000	$+6,000$
C	26,000	100%	26,000	26,000	0
D	48,000	10%	4,800	6,000	$+1,200$
E	56,000	20%	11,200	20,000	$+8,800$
F	30,000	20%	6,000	4,000	$-2,000$
G	80,000	0%	0	0	0
H	16,000	0%	0	0	0
Total	308,000		100,000	112,000	$+12,000$

Table: Monitoring and Controlling Budgeted Cost (Overrun)

Note

Overrun

Project crashing

- Projects will sometimes have deadlines that are impossible to meet using normal procedures
- By using exceptional methods, it may be possible to finish the project in less time
- Costs usually increase
- Reducing a project's completion time is called crashing

- Start by using the normal time to create the critical path
- The normal cost is the cost for completing the activity using normal procedures
- If the project will not meet the required deadline, extraordinary measures must be taken
  - The crash time is the shortest possible activity time and will require additional resources
  - The crash cost is the price of completing the activity in the earlier-than-normal time

Four Steps of Project Crashing

1. Find the normal critical path and identify the critical activities.
2. Compute the crash cost per week (or other time period) for all activities in the network. This process uses the following formula:[4]

$$\text{Crash cost/Time period} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}} \quad \textbf{(11-10)}$$

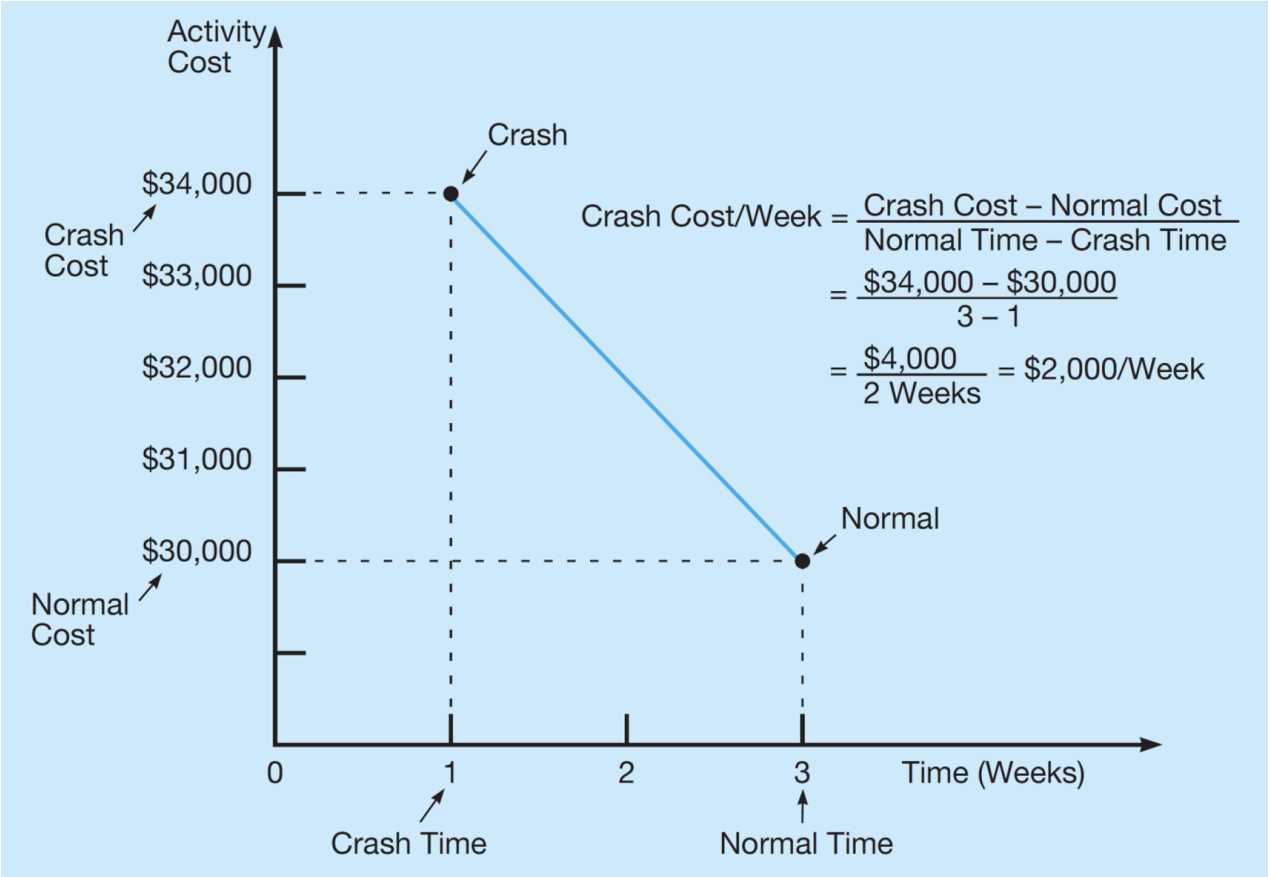
3. Select the activity on the critical path with the smallest crash cost per week. Crash this activity to the maximum extent possible or to the point at which your desired deadline has been reached.
4. Check to be sure that the critical path you were crashing is still critical. Often, a reduction in activity time along the critical path causes a noncritical path or paths to become critical. If the critical path is still the longest path through the network, return to step 3. If not, find the new critical path and return to step 3.

Project crashing: General Foundry

TABLE 11.9 Normal and Crash Data for General Foundry, Inc.

ACTIVITY	TIME (WEEKS)		COST ($)		CRASH COST PER WEEK ($)	CRITICAL PATH?
	NORMAL	CRASH	NORMAL	CRASH		
A	2	1	22,000	23,000	1,000	Yes
B	3	1	30,000	34,000	2,000	No
C	2	1	26,000	27,000	1,000	Yes
D	4	3	48,000	49,000	1,000	No
E	4	2	56,000	58,000	1,000	Yes
F	3	2	30,000	30,500	500	No
G	5	2	80,000	86,000	2,000	Yes
H	2	1	16,000	19,000	3,000	Yes

FIGURE 11.11 Crash and Normal Times and Costs for Activity B

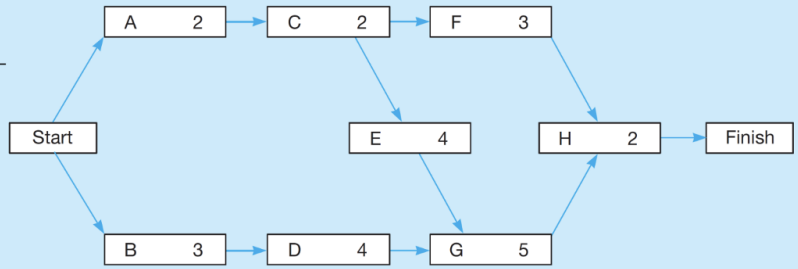


Project crashing: General Foundry

TABLE 11.9 Normal and Crash Data for General Foundry, Inc.

ACTIVITY	TIME (WEEKS)		COST ($)		CRASH COST PER WEEK ($)	CRITICAL PATH?
	NORMAL	CRASH	NORMAL	CRASH		
A	2	1	22,000	23,000	1,000	Yes
B	3	1	30,000	34,000	2,000	No
C	2	1	26,000	27,000	1,000	Yes
D	4	3	48,000	49,000	1,000	No
E	4	2	56,000	58,000	1,000	Yes
F	3	2	30,000	30,500	500	No
G	5	2	80,000	86,000	2,000	Yes
H	2	1	16,000	19,000	3,000	Yes

FIGURE 11.12 General Foundry's Network with Activity Times



Questions

